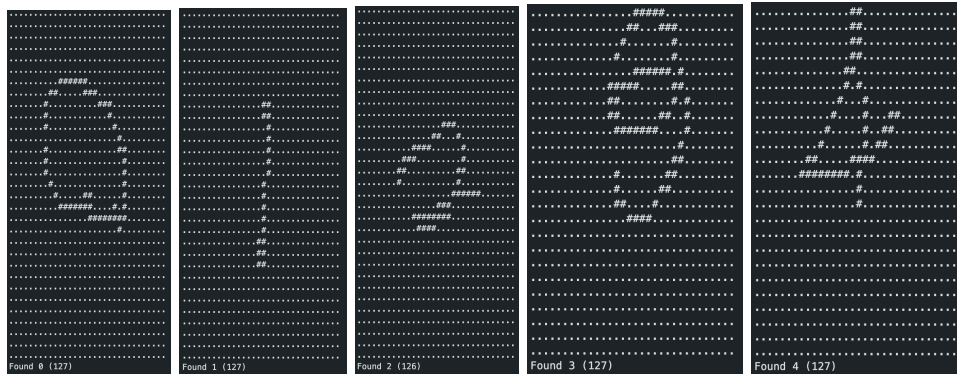


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Fig. 1 Screenshot of the Serial Monitor outputs for digits 0-9 before fine-tuning



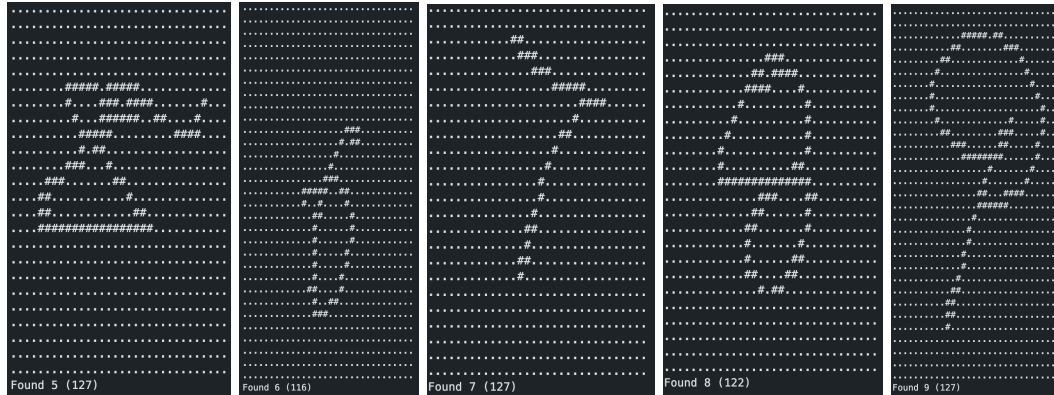


Fig. 2 Screenshot of the Serial Monitor outputs for digits 0-9 after fine-tuning

Reflection

The fine-tuned model performed significantly better than the baseline model for my own gesture data. Before fine-tuning, the model was less consistent because the baseline model was trained on general gesture patterns that did not fully match my drawing style. Some digits were misclassified, especially digits with similar shapes such as 3, 4, 5, and 7. After fine-tuning with my collected gesture data, the model became much more accurate and more stable when recognizing my gestures.

The fine-tuned model achieved an accuracy of 91% on my collected dataset. Most digits were classified correctly, and digits such as 1 and 2 reached perfect precision and recall. The remaining errors mainly occurred between visually similar gestures. For example, one 0 was classified as 9, one 6 was classified as 7, and some 9 samples were confused with 5 or 8. This makes sense because these digits can look similar when drawn in the air using motion gestures.

Overall, the fine-tuned model performed better than the baseline model because it was trained using my own gesture style, including my drawing speed, direction, and motion pattern. Fine-tuning helped the model adapt to the way I personally draw each digit, so it improved the recognition performance for this specific user. This shows that personalization is important for gesture-recognition TinyML applications, especially when different users may draw the same digit in slightly different ways.